

Nginx Multi-Site Setup (Static Websites on Different Ports)

This guide sets up Nginx to serve two independent static websites, each on its own port, on a single Linux server.

- Site 1 → `http://localhost:8081`
 - Site 2 → `http://localhost:8082`
-

1. Install Nginx

Ubuntu / Debian

```
sudo apt update
sudo apt install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

CentOS / RHEL / Fedora

```
sudo dnf install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

Check status:

```
systemctl status nginx
```

2. Create the Website Files

```
sudo mkdir -p /var/www/site1
sudo mkdir -p /var/www/site2

echo "<h1>Site 1 - Running on port 8081</h1>" | sudo tee
/var/www/site1/index.html
echo "<h1>Site 2 - Running on port 8082</h1>" | sudo tee
/var/www/site2/index.html
```

Set ownership:

```
# Ubuntu/Debian
sudo chown -R www-data:www-data /var/www/site1 /var/www/site2

# CentOS/RHEL
sudo chown -R nginx:nginx /var/www/site1 /var/www/site2
```

3. Create Server Block (vhost) Configs

Ubuntu / Debian

Config location: /etc/nginx/sites-available/

/etc/nginx/sites-available/site1

```
server {
    listen 8081;
    server_name localhost;
    root /var/www/site1;
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
```

/etc/nginx/sites-available/site2

```
server {
    listen 8082;
    server_name localhost;
    root /var/www/site2;
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
```

Enable them:

```
sudo ln -s /etc/nginx/sites-available/site1 /etc/nginx/sites-enabled/
sudo ln -s /etc/nginx/sites-available/site2 /etc/nginx/sites-enabled/
```

CentOS / RHEL / Fedora

Config location: /etc/nginx/conf.d/

Create /etc/nginx/conf.d/site1.conf and /etc/nginx/conf.d/site2.conf with the same server { } blocks as above (no sites-enabled symlink step needed — files in conf.d/ are picked up automatically).

4. Test and Reload Nginx

```
sudo nginx -t          # Validates config syntax
sudo systemctl reload nginx
```

5. Open Firewall Ports

```
# UFW (Ubuntu)
sudo ufw allow 8081/tcp
sudo ufw allow 8082/tcp
```

```
# firewallld (CentOS/RHEL)
sudo firewall-cmd --permanent --add-port=8081/tcp
sudo firewall-cmd --permanent --add-port=8082/tcp
sudo firewall-cmd --reload
```

SELinux (CentOS/RHEL only, if enforcing)

```
sudo semanage port -a -t http_port_t -p tcp 8081
sudo semanage port -a -t http_port_t -p tcp 8082
```

6. Verify

```
curl http://localhost:8081
curl http://localhost:8082
```

Or open in a browser: - `http://<server-ip>:8081` - `http://<server-ip>:8082`

Directory Structure Summary

<code>/var/www/site1/index.html</code>	→ served on port 8081
<code>/var/www/site2/index.html</code>	→ served on port 8082
<code>/etc/nginx/sites-available/site1</code>	→ config (Debian/Ubuntu)
<code>/etc/nginx/sites-available/site2</code>	→ config (Debian/Ubuntu)
<code>/etc/nginx/sites-enabled/site1</code>	→ symlink (Debian/Ubuntu)
<code>/etc/nginx/sites-enabled/site2</code>	→ symlink (Debian/Ubuntu)
<code>/etc/nginx/conf.d/site1.conf</code>	→ config (CentOS/RHEL)
<code>/etc/nginx/conf.d/site2.conf</code>	→ config (CentOS/RHEL)

Adding a Third Site

1. Create a new directory under `/var/www/site3` with an `index.html`.
2. Copy an existing server block config, rename it `site3`, and change listen to a free port (e.g. 8083) and root to `/var/www/site3`.
3. Enable it (symlink on Debian/Ubuntu, or drop the file in `conf.d/` on CentOS/RHEL).
4. Run `sudo nginx -t && sudo systemctl reload nginx`.
5. Open the new port in the firewall.

See `TROUBLESHOOTING.md` for common issues and fixes.